Chiral polyhedra of order twice an odd prime power

The human_compression project

Topic F: second application, private draft

Abstract

We recover the three main theorems of Kong, Feng, Hou, Leemans and Qu for groups of order $2p^m$: the structural and tightness characterization, tightness at $m = 3, 4$, and the complete stated non-tight constructions. The common cyclotomic lattice supplies exact presentations for every parameter. An elementary small-kernel argument includes the prime three without a classification table. We also audit the regular byproducts: three asserted families exist for every odd prime, while the fourth exists exactly for the prime three. The previously proved cyclic lift is included with its separate attribution and scope.

1 The structural theorem

We retain the classical rank-three rotation-group correspondence: a finite generating pair (a, b) of orders at least three satisfying $(ab)^2 = 1$ and $\langle a\rangle \cap \langle b\rangle = 1$ determines a directly regular or chiral polyhedron. It is chiral exactly when no automorphism simultaneously inverts a, b. In the chiral case the full group is $G = \langle a, b\rangle$, and tightness means $|G| = |a||b|$. This statement is the exact geometric interface used in the source's Section 2.2, attributed there to Schulte and Weiss; it is not newly proved by the following group calculations.

Theorem 1 (Source Theorem 1.1)**.** *Let $(G; a, b)$ be the full group of a chiral polyhedron and $|G| = 2p^m$ for an odd prime p. After exchanging a, b if necessary,*

$$|a| = p^u, \quad |b| = 2p^v, \quad u, v \geq 1, \qquad G = P \rtimes \langle ab\rangle, \quad P = \langle a, b^2\rangle.$$

The Sylow subgroup P is nonabelian and needs exactly two generators. The polyhedron is tight if and only if P is metacyclic.

Proof. The Sylow subgroup P has index two and is normal. The product ab is a nonidentity involution: equality to one would contradict the nontrivial disjoint cyclic subgroups. Thus $\langle ab\rangle$ meets P trivially, giving the displayed semidirect product. Both generators cannot be in P, because they generate G. Neither can both be outside it, because their product would be an involution in the odd-order group P. Thus exactly one is outside; orders and nondegeneracy give the displayed type. Put $x = a$ and $y = b^2$. The identities

$$x^b = y^{-1}x^{-1}, \qquad y^b = y$$

show $\langle x, y\rangle$ is normalized by a, b. Its quotient in G is generated by the image of b, of order at most two. It is contained in P, so equals P. If P were cyclic, the two nontrivial cyclic subgroups $\langle x\rangle, \langle y\rangle$ would share its unique subgroup of order p, contradicting the intersection condition. Hence two generators are necessary as well as sufficient. If P were abelian, the abelian-kernel reversal lemma of [1] with $t = ab$ would supply a reversal, contradicting chirality. In particular $m \geq 3$.

For the last assertion we retain precisely two classical odd-primary subgroup facts, also invoked as source Theorems 2.4 and 2.5: (i) a finite odd p-group which is a setwise product of two cyclic subgroups is metacyclic (Huppert, *Endliche Gruppen I*, Theorem 11.5 as cited in the source); (ii) in a

finite metacyclic odd p-group, subgroups permute (the Iwasawa result cited through Schmidt, *Subgroup Lattices of Groups*, Theorem 2.3.1). These are retained classical inputs with full proof cost unknown, not deductions from the finite diagnostics. The two cyclic subgroups here intersect trivially. If the polyhedron is tight, their setwise product has size $p^{u+v} = p^m = |P|$; it is P, so (i) applies. Conversely (ii) makes their setwise product a subgroup containing the two generators of P. It is therefore P, and the order formula gives tightness. □

2 Tightness at the two smallest nonabelian orders

Theorem 2 (Uniform small-order reversal)**.** *Let $G = \langle a, b\rangle$ have order $2p^m$, where $m = 3$ or 4. Assume $|a|, |b| \geq 3$, $(ab)^2 = 1$, $\langle a\rangle \cap \langle b\rangle = 1$ and $|a||b| < |G|$. Then G has an automorphism simultaneously inverting a, b. In particular every chiral polyhedron of either order is tight.*

Proof. The Sylow subgroup P of odd order is normal of index two. Exactly one of a, b is in P: neither both inside (generation) nor both outside (their nonidentity involution product would be inside). Exchange them if necessary and put $x = a$, $y = b^2$. Conjugation by b satisfies $x^b = y^{-1}x^{-1}$ and $y^b = y$, so $P = \langle x, y\rangle$. Their cyclic subgroups are nontrivial and disjoint. If P were abelian, its independent cyclic coordinates give an inverting automorphism of the presentation

$$\langle a, b \mid a^{|a|},\ b^{|b|},\ (ab)^2,\ [a, b^2]\rangle :$$

collection bounds its order by $|a||b|$, contradicting non-tightness. Thus P is nonabelian. Write $|x| = p^u$, $|y| = p^v$. Disjointness and non-tightness give $u, v \geq 1$, $u + v < m$.

For $m = 3$ we have $u = v = 1$ and class two. With $c = [x, y]$, the relations $x^p = y^p = c^p = [c, x] = [c, y] = 1$ collect words into $x^i y^j c^k$, at most p^3 possibilities. Adjoin b with $b^2 = y$ and $(xb)^2 = 1$. The subgroup generated by x, b^2 is normal by the conjugation formula and has index at most two. The resulting presented group maps onto G, and the bound $2p^3$ shows it is exactly G. Simultaneous inversion of a, b respects all relations: it inverts x, y, fixes their commutator up to conjugation, and preserves class two. The resulting endomorphism squares to the identity and is the required automorphism.

Let $m = 4$. The possible pairs (u, v) are $(1, 1), (1, 2), (2, 1)$. If the class is two, P' is cyclic, generated by $[x, y]$; the small-kernel lemma of [1] therefore gives $|P'| = p$. The pair $(1, 1)$ would bound the order by p^3 and is impossible. For the other two pairs, the relations

$$x^{p^u} = y^{p^v} = c^p = [c, x] = [c, y] = 1$$

collect into at most $p^{u+v+1} = p^4$ words. Exactly the preceding presentation and reversal argument applies.

Suppose the class is three. Put $c = [x, y]$, $d = [c, x]$, $e = [c, y]$. Impose all weight-four commutators equal to one, and $c^p = d^p = e^p = 1$. These are actual relations by the small-kernel lemma of [1]. If $u = v = 1$, the two elements d, e generate the cyclic group $\gamma_3(P)$, so there is a relation $d^i e^j = 1$ with $(i, j) \neq (0, 0)$ modulo p. The corresponding presentation of P has at most p^4 collected words $x^\alpha y^\beta c^\gamma d^\delta e^\epsilon$: the last two exponents have only p independent possibilities. If $(u, v) = (2, 1)$, then x^p is central, since $[x^p, y] = c^p d^{\binom{p}{2}} = 1$. It is nontrivial, so generates $Z(P) = \gamma_3(P)$. Thus $d = x^{pi}$ and $e = x^{pj}$ for some i, j; collection into $x^\alpha y^\beta c^\gamma$ gives the bound p^4. For $(u, v) = (1, 2)$ interchange x, y in this central-power argument.

In each case these full relations, together with $b^2 = y$ and $(xb)^2 = 1$, present G by the same surjection and order bound. Under $a \mapsto a^{-1}$, $b \mapsto b^{-1}$ we have $x \mapsto x^{-1}$, $y \mapsto y^{-1}$. In a class-three group their new commutator is c times an element of the central group $\langle d, e\rangle$, so still has pth power one. The new triple commutators are d^{-1}, e^{-1}. Consequently the linear relation in the $(1, 1)$ case is preserved, and in the other cases the two equations with x^p or y^p are both inverted. Nilpotency class

at most three is preserved under any substitution, as are the orders and $(ab)^2 = 1$. This proves the required involutory automorphism for every odd p, including $p = 3$, without appealing to a finite catalogue. □

3 Every odd exponent at least five

Theorem 3 (Source Theorem 1.4)**.** *For every odd prime p and every odd $m \geq 5$, and for every even $m \geq p + 3$, there is a non-tight chiral polyhedron of order $2p^m$. The following two constructions realize the exact source presentations and have type $\{p^2, 2p^e\}$, with e specified below.*

The proof is the complete construction in this section and the next.

First construct the underlying source p-group for every $n = m-1 \geq 2$, without a parity restriction. Let $O = \mathbb{Z}[\zeta]$, $\pi = \zeta - 1$, where ζ is a primitive pth root, and write $n = (e-1)(p-1)+r$, $1 \leq r \leq p-1$. Set

$$M = O/\pi^n O, \qquad s = 1, \qquad Tv = \zeta v, \qquad z = p^{e-1}\pi^{r-1}.$$

The common lattice theorem gives $|M| = p^n$, $|s| = p^e$ and $0 \neq z \in \pi^{n-1}O/\pi^n O$, so $|z| = p$ and $Tz = z$. The cyclic-extension construction in the proof of the carry theorem, which needs only $T^p = 1$ and $Tz = z$ to construct P, gives

$$P = (M \rtimes \mathbb{Z})/\langle(-z, p)\rangle, \qquad |P| = p^{n+1}, \qquad v^\beta = Tv, \quad \beta^p = z.$$

The powers of $\zeta = 1 + \pi$ span M, so $P = \langle\beta, s\rangle$. As M is abelian, $[P, P] = \pi M$ and inductively $\gamma_j(P) = \pi^{j-1}M$ for every $j \geq 2$. Thus $\gamma_n(P) \neq 1$ and $\gamma_{n+1}(P) = 1$: its class is n. The norm $\sum_{i=0}^{p-1} T^i$ is zero, so $(\beta s)^p = \beta^p = z$ and $|\beta s| = p^2$. More generally $(\beta^k v)^p = kz$ for $1 \leq k < p$ and $v \in M$, since multiplication by k permutes the powers of T. Every element outside M therefore has order p^2. An automorphism fixing β and inverting s would invert all its β-conjugates, hence all of M, while fixing $z = \beta^p$. This contradicts $|z| = p > 2$. These conclusions require no reflection S.

For the literal source presentation, put $s_i = \pi^{i-1}$. The mixed coordinate orders are precisely those printed in Section 3; $[s_i, \beta] = s_{i+1}$ and

$$[s_{p-1}, \beta] = \pi^{p-1} = -\sum_{i=1}^{p-1} \binom{p}{i} \pi^{i-1}.$$

The carry is $\beta^p = s_r^{p^{e-1}}$. Conversely those relations collect every source word into $\beta^j \prod s_i^{a_i}$ with $0 \leq j < p$ and the indicated coordinate ranges, including coordinates of order one. They bound the presented order by p^{n+1}, and the constructed model realizes equality. This proves the exact presentation for every $n \geq 2$; it does not assert the classification of all maximal-class groups.

For the chiral construction now assume that $n \geq 4$ is even. Then r is even and $r \geq 2$. Put $Sv = \bar{v}$. Conjugation satisfies $\bar{\pi} = -\zeta^{-1}\pi$, hence $Sz = (-1)^{r-1}z = -z$, and $STS = T^{-1}$ and $Ss = s$. In the direct coordinates $1, \pi, \ldots, \pi^{p-2}$, z is a nonzero element of the rth coordinate whereas s is in the first; thus $\langle z\rangle \cap \langle s\rangle = 0$. The carry theorem of [1] gives order $2p^{n+1}$ and type $\{p^2, 2p^e\}$. This is non-tight since $n + 1 > e + 2$ for even $n \geq 4$ and $p \geq 3$.

The two source formulas for conjugation are UF.Conjugation, which also explains the long binomial verification without retaining it as an external proof premise.

4 The literal even-exponent construction

Keep even $n \geq 4$, now with $e \geq 2$, and write $A = O/\pi^n O$. Replace its additive module by

$$M = \{(v, j) \in A \times \mathbb{Z}/p^2\mathbb{Z} : v \bmod \pi = j \bmod p\}.$$

Set $s = (1, 1)$, $z = (p^{e-1}\pi^{r-1}, 0)$ and $T(v, j) = (\zeta v, j)$, $S(v, j) = (\bar{v}, j)$. Then $|M| = p^{n+1}$, $|s| = p^e$ and all the involution and carry conditions of the cyclic-carry theorem of [1] hold. The subgroup generated by the T-orbit projects onto A. It contains

$$\sum_{i=0}^{p-1} T^i s = (0, p),$$

which generates the projection kernel. It is therefore all of M. Here $w = (p^{e-1}\pi^{r-1}, p)$ has order p. Projection to A shows $\langle w\rangle \cap \langle s\rangle = 0$, since the nonzero first coordinate of w is in coordinate $r \geq 2$. Thus the group has order $2p^{n+2}$, type $\{p^2, 2p^e\}$ and is chiral and non-tight. These are exactly all the source's even exponents $m \geq p + 3$: $n = m - 2 \geq p + 1$ is even, so $e \geq 2$.

Set $s_1 = s$, $s_i = (\pi^{i-1}, 0)$ for $i \geq 2$, and $x = (0, -p)$. The cyclic norm identity gives exactly the source relation

$$x = -\sum_{i=1}^{p-1} \binom{p}{i} s_i - [s_{p-1}, \beta].$$

It also gives x central of exact order p; $Sx = x$. The source quotient by $\langle x\rangle$ is the preceding odd-exponent presentation, so its order is at most $p \cdot 2p^{n+1}$. The displayed model has this order and satisfies every source relation, proving equality and the exact presentation bridge.

There is a minor correction to the printed lower-bound argument. The proposed map in Section 6, Claim 4 to $C_{p^2} \times C_p$ sends σ to 1 and β to a nonidentity element of order p, so violates $\beta^\sigma = \beta^{-1}$. The map to C_{p^2} sending s_1 to a generator and β, σ to 1 does respect the relations and sends x to a nonzero pth power. This suffices for that lower bound; the full model above proves it directly.

5 Two markings on one cyclotomic group

For an odd prime p and $n \geq 2$, let $A = \mathbb{Z}[\zeta_p]/(\zeta_p - 1)^n$, $P = A \rtimes \langle\beta\rangle$, with $v^\beta = \zeta_p v$ and $|\beta| = p$. Let σ act by conjugation on A and inversion on β; let τ negate A and fix β. The shared split cyclotomic construction of [1] proves that $R = P \rtimes \langle\sigma, \tau\rangle$ has order $4p^{n+1}$ and the marking

$$(s\tau\sigma,\ \beta\tau\sigma,\ \sigma), \qquad s = 1 \in A,$$

of type $\{p, 2p\}$. The complete construction and its proof are included in the common paper; the first application is not an additional proof premise. Taking $n = 2, 3$ gives respectively the first two families in source Remark 1.3 for every odd prime.

There is a second marking on the same group. In $G = P \rtimes \langle\sigma\rangle$ put $a = \beta s$ and $b = s^{-1}\sigma$. The norm identity gives $|a| = p$; and $|b| = 2|s| = 2p^e$, where $n = (e - 1)(p - 1) + r$, $1 \leq r \leq p - 1$. Their product is $\beta\sigma$, an involution. They generate G: $b^2 = s^{-2}$ recovers s, then β and σ. Their cyclic subgroups intersect trivially: a common element lies in $P \cap \langle b\rangle = \langle s\rangle$, whereas every nonidentity power of a has nonzero image in P/A. The element $c = \tau\sigma s$ in R is an involution and lies outside G. It fixes σ by conjugation, inverts s, and sends β to $s^{-1}\beta^{-1}s$. Consequently it simultaneously inverts a, b. The rotation-pair argument of [1] gives a second regular marking of R of type $\{p, 2p^e\}$. For $p = 3$, $n = 3$, this supplies the fourth claimed type $\{3, 18\}$ at order $324 = 4 \cdot 3^4$.

6 A mixed Heisenberg construction

For every odd prime p, define H by triples

$$(i, j, k) \in \mathbb{Z}/p^2\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z},$$

with multiplication

$$(i, j, k)(i', j', k') = (i + i', j + j', k + k' - i'j).$$

The formula is associative because its final correction is bilinear; it has an identity and inverses. Set $x = (1,0,0)$, $v = (0,1,0)$, $c = (0,0,1)$. Then $[x,v] = c$, c is central, and

$$|H| = p^4, \quad |x| = p^2, \quad |v| = |c| = p, \quad H = \langle x, v \rangle.$$

In particular this explicitly realizes the presentation $x^{p^2} = v^p = c^p = [c,x] = [c,v] = 1$, $[x,v] = c$; its normal forms give the matching upper bound. Define

$$\theta(i,j,k) = (-i, j-i, -i(i+1)/2 - k),$$

where the last two coordinates are taken modulo p. The value is independent of the representative of i, and direct substitution in the bilinear multiplication proves it is an automorphism. Alternatively the exact presentation shows it sends $x \mapsto v^{-1}x^{-1}$, $v \mapsto v$, $c \mapsto c^{-1}$. Its square is $(i,j,k) \mapsto (i,j,k+i)$, conjugation by v.

In the semidirect product of H by an infinite cyclic generator b acting by $h^b = \theta(h)$, the element b^2v^{-1} is central: θ^2 is conjugation by v and $\theta(v) = v$. Quotient by its cyclic subgroup. The intersection of that subgroup with H is trivial, as its nontrivial powers have nonzero integer coordinate. The quotient G therefore contains H with index two and satisfies $b^2 = v$; hence $|G| = 2p^4$ and $|b| = 2p$. The relation $x^b = v^{-1}x^{-1}$ gives $(xb)^2 = 1$. Moreover $G = \langle x, b \rangle$ and $\langle x \rangle \cap \langle b \rangle = 1$, since the intersection is inside $\langle x \rangle \cap \langle v \rangle = 1$ in the explicit coordinates. The presentation is invariant under $x \mapsto x^{-1}$, $b \mapsto b^{-1}$: it inverts $v = b^2$, preserves the central commutator of order p and all displayed orders. The resulting map is involutory. The common rotation-pair argument constructs a regular polyhedron of order $4p^4$ and type $\{p^2, 2p\}$. This supplies the third family without a group catalogue.

7 Why the fourth family has only the prime three

For $p \geq 5$, suppose a rank-three string C-group R had order $4p^4$ and type $\{p, 2p^2\}$. Its Sylow p-subgroup P is normal: its number divides four and is one modulo p. The commuting endpoints generate a Klein four complement, so $R/P \cong C_2^2$. Put $a = r_0r_1$ and $b = r_1r_2$. Since $|a| = p$ is odd, $r_0P = r_1P$, whereas r_2P is the other independent endpoint image. The rotation subgroup $G = \langle a, b \rangle$ is normal of index at most two, because r_1 inverts both its generators and $R = \langle G, r_1 \rangle$. Its image in R/P has order two, so its index is exactly two. Thus $|G| = 2p^4$.

As in the structural proof, $P = \langle x, y \rangle$ for $x = a$ and $y = b^2$. It is nonabelian: an abelian group generated by these disjoint cyclic subgroups would have order only p^3. The elementary small-kernel lemma in the common paper says that P' has exponent p and P has class at most three. For $p \geq 5$, collection gives $(uv)^p = u^pv^p$ in P. But x and y have orders p, p^2, and $x^b = y^{-1}x^{-1}$. Taking pth powers contradicts $|y| = p^2$. Thus the fourth family of Remark 1.3 does not exist for $p \geq 5$; the preceding cyclotomic example proves existence at $p = 3$. This corrects an auxiliary printed assertion, not a numbered open problem, and makes no priority claim.

8 The previously handled small even range

For even $6 \leq m \leq p+1$, put $n = m-2$. The odd construction above has $e = 1$, $r = n$ and gives G_0 of order $2p^{m-1}$ with type $\{p^2, 2p\}$. Residue modulo π on the module, with β, σ sent to zero, defines a surjection to the additive group $\mathbb{F}_p$. The carry has zero residue because $r \geq 4$; both T and conjugation preserve residue. Thus all presentation relations are respected. It sends $a_0 = \beta s$ to 1 and $b_0 = s^{-1}\sigma$ to -1. The common cyclic-lift theorem gives order $2p^m$ and type $\{p^2, 2p^2\}$. Its type product is $2p^4 < 2p^m$. Together with the two preceding constructions this covers every $m \geq 5$. The small-order theorem proves the complementary nonexistence for non-tight chiral polyhedra at $m < 5$.

This paragraph reprises the previously reviewed central-lift result. The accepted source's Section 7 literally uses the word "tight" when discussing its missing even range; the surrounding discussion

and Theorem 1.4 concern non-tight examples, while its introduction already cites tight examples. We distinguish that literal wording from the non-tight existence conclusion proved here. There is no new numbered-conjecture credit or claim about current priority in this edition. The source's tight existence result from Cunningham remains attributed background rather than a construction silently claimed here.

9 Exact source scope and retained inputs

The source edition is the complete 22-page accepted manuscript of [2], accepted 19 January 2026. The structural theorem recovers its Theorem 1.1, the uniform reversal theorem recovers Theorem 1.2, and the odd/even constructions recover both cases of Theorem 1.4. The cyclic-carry presentation comparison includes the precise Section 3 generators, their mixed orders, conjugation formulas and the full Section 6 central extension. The maximal-class assertion follows directly: in the odd family the lower-central terms from the second onward are $\pi A, \pi^2 A, \ldots$, so the class is n for a group of order p^{n+1}.

The four clauses of Remark 1.3 are all decided with explicit models or an obstruction. Its fourth clause requires the prime-three qualification proved above. The printed Section 6 map to $C_{p^2} \times C_p$ is repaired by the explicit module model and the valid cyclic quotient. In the small-order proof we impose every weight-four relation and use a nonzero dependence of the two central triple commutators; we do not rely on the source's asymmetric presentation-order shortcut. None of these qualifications is reported as a new conjecture solution.

The only source-specific classical classification inputs retained in this application are the two odd-primary subgroup facts stated at the point of use. Their original books are not retained as complete local PDFs, and their external proof costs remain unknown. The first application retains its separate Cunningham–Pellicer input unchanged. The source's small-order group classification, Conder list and maximal-class existence theorem are not premises of the proofs here. Introductory census percentages, survey history and tight examples cited from Cunningham are not regenerated.

All finite programs are validation of the uniform written proofs. Their complete files, including the small-group exploratory scan when retained, must be inventoried; a successful finite scan is not substituted for a quantified theorem. The generic formal statements are scoped in the common paper. There is no full Lean proof of any source main theorem, no new source admission in this private draft, and no numerical compression claim.

References

[1] The human_compression project, *Odd kernels, cyclotomic modules and cyclic carries*, accompanying common paper.

[2] T.-T. Kong, Y.-Q. Feng, D.-D. Hou, D. Leemans and H.-P. Qu, *Chiral 3-polytopes of order* $2p^m$, accepted manuscript, 19 January 2026, DOI10.26493/1855-3974.3716.b64.